

Vector Calculus & Probability Dissection: SPECTRE AI Timeline

Note. This document translates the chronological $T = 0 \rightarrow T = 7$ execution timeline into strict mathematical formulations, tracking tensor dimensionality, sampling, and probabilistic constraints throughout the pipeline.

1 Phase 1: Live Forward Pass (Inference Math)

$T = 0$: Deterministic Intake (Sensor Fusion)

Let

$$I_{\text{img}} \in \mathbb{R}^{4 \times 64 \times 64}, \quad I_{\text{tel}} \in \mathbb{R}^D. \quad (1)$$

The image and telemetry encoders produce

$$E_{\text{img}} = f_{\text{CNN}}(I_{\text{img}}) \in \mathbb{R}^{1 \times 128}, \quad (2)$$

$$E_{\text{tel}} = f_{\text{MLP}}(I_{\text{tel}}) \in \mathbb{R}^{1 \times 64}. \quad (3)$$

Define concatenated features

$$\text{Concat}_t = [E_{\text{img}} \parallel E_{\text{tel}}] \in \mathbb{R}^{1 \times 192}. \quad (4)$$

The gate is projected to the same dimensionality:

$$G = \sigma(\text{Concat}_t W_g + b_g) \in [0, 1]^{1 \times 192}. \quad (5)$$

Element-wise fusion is then

$$F_t = G \odot \text{Concat}_t \in \mathbb{R}^{1 \times 192}. \quad (6)$$

At $T = 0$, variance is fixed at $\sigma^2 = 0$, so this stage is deterministic.

$T = 1$: PEARL Context z (Variational Belief)

Given history matrix $H \in \mathbb{R}^{16 \times 64}$, the causal GRU produces

$$c_t \in \mathbb{R}^{1 \times 64}. \quad (7)$$

Gaussian parameters for latent belief:

$$\mu_z = c_t W_\mu + b_\mu \in \mathbb{R}^{1 \times 32}, \quad (8)$$

$$\log \sigma_z^2 = c_t W_\sigma + b_\sigma \in \mathbb{R}^{1 \times 32}. \quad (9)$$

Reparameterized sample:

$$z_t = \mu_z + \exp\left(\frac{1}{2} \log \sigma_z^2\right) \odot \epsilon_z, \quad \epsilon_z \sim \mathcal{N}(0, I_{32}). \quad (10)$$

$T = 2$: RSSM Prior Target (Physics Dreaming)

The deterministic recurrent core state is

$$h_t \in \mathbb{R}^{1 \times 256}. \quad (11)$$

Prior parameters are predicted without direct access to F_t :

$$[\mu_{\text{prior}}, \sigma_{\text{prior}}] = \text{Linear}(h_t) \in \mathbb{R}^{1 \times 128} \rightarrow \mathbb{R}^{1 \times 64} \oplus \mathbb{R}^{1 \times 64}. \quad (12)$$

Prior sample:

$$\hat{z}_{\text{prior}} = \mu_{\text{prior}} + \sigma_{\text{prior}} \odot \epsilon_p \in \mathbb{R}^{1 \times 64}. \quad (13)$$

$T = 3$: Action Sample and Jacobian (Motor Policy)

Actor input:

$$X_t = [F_t, z_t] \in \mathbb{R}^{1 \times 224}. \quad (14)$$

Policy parameters:

$$\mu_a, \sigma_a = f_{\text{Actor}}(X_t) \in \mathbb{R}^{1 \times 3}. \quad (15)$$

Unbounded sample:

$$u_t = \mu_a + \sigma_a \odot \epsilon_a \in \mathbb{R}^{1 \times 3}. \quad (16)$$

Bounded action and log-density correction:

$$a_t = \tanh(u_t), \quad (17)$$

$$\log \pi(a_t | s_t) = \sum_{i=1}^3 \left[\log \mathcal{N}(u_{t,i} | \mu_{a,i}, \sigma_{a,i}^2) - \log(1 - \tanh^2(u_{t,i})) \right] \in \mathbb{R}^{1 \times 1}. \quad (18)$$

In implementation, the Jacobian term is evaluated in numerically stable form:

$$\log(1 - \tanh^2(u)) = 2(\log 2 - u - \text{softplus}(-2u)), \quad (19)$$

to avoid $\log(0)$ under finite-precision saturation when $|u|$ is large.

Manual Simulation of Phase 1 ($T = 0$ to $T = 3$)

This subsection provides an illustrative numeric forward-pass trace for a concrete scenario: the agent approaches a sharp right turn at approximately 200 km/h.

Step $T = 0$: Sensor Capture and Fusion. Assume representative scalar components from the encoded vectors are

$$E_{\text{img}}^{(1)} = 0.800, \quad E_{\text{tel}}^{(1)} = 0.900, \quad (20)$$

and gate value

$$G^{(1)} = 0.800. \quad (21)$$

Then the fused feature component is

$$F_t^{(1)} = G^{(1)} \odot E_{\text{img}}^{(1)} + (1 - G^{(1)}) \odot E_{\text{tel}}^{(1)} = (0.800 \odot 0.800) + (0.200 \odot 0.900) = 0.820. \quad (22)$$

Step $T = 1$: PEARL Belief Sampling. Assume

$$\mu_z^{(1)} = 0.500, \quad \sigma_z^{(1)} = 0.100, \quad \epsilon_z^{(1)} \sim \mathcal{N}(0, 1), \quad \epsilon_z^{(1)} = 1.200. \quad (23)$$

Reparameterized latent sample:

$$z_t^{(1)} = \mu_z^{(1)} + \sigma_z^{(1)} \odot \epsilon_z^{(1)} = 0.500 + (0.100 \odot 1.200) = 0.620. \quad (24)$$

Step $T = 2$: RSSM Prior Prediction. The prior branch produces $(\mu_{\text{prior}}, \sigma_{\text{prior}})$ and sample $\hat{z}_{\text{prior}}$ for later consistency matching at $T = 5$.

Step $T = 3$: Actor Sampling and Squashing. Using fused/contextual inputs (including $F_t^{(1)} = 0.820$ and $z_t^{(1)} = 0.620$), let

$$\mu_a = [0.600, 1.200, -2.000], \quad (25)$$

$$\sigma_a = [0.200, 0.050, 0.010], \quad (26)$$

$$\epsilon_a \sim \mathcal{N}(0, I_3), \quad \epsilon_a = [1.500, -0.800, 0.500]. \quad (27)$$

Raw action vector:

$$u_t = \mu_a + \sigma_a \odot \epsilon_a = \begin{bmatrix} 0.600 \\ 1.200 \\ -2.000 \end{bmatrix} + \left(\begin{bmatrix} 0.200 \\ 0.050 \\ 0.010 \end{bmatrix} \odot \begin{bmatrix} 1.500 \\ -0.800 \\ 0.500 \end{bmatrix} \right) = \begin{bmatrix} 0.900 \\ 1.160 \\ -1.995 \end{bmatrix}. \quad (28)$$

Component-wise:

$$u_{\text{steer}} = 0.600 + (0.200 \times 1.500) = 0.900, \quad (29)$$

$$u_{\text{gas}} = 1.200 + (0.050 \times -0.800) = 1.160, \quad (30)$$

$$u_{\text{brake}} = -2.000 + (0.010 \times 0.500) = -1.995. \quad (31)$$

After physical squashing $a_t = \tanh(u_t)$:

$$a_{\text{steer}} = \tanh(0.900) \approx 0.716, \quad (32)$$

$$a_{\text{gas}} = \tanh(1.160) \approx 0.821, \quad (33)$$

$$a_{\text{brake}} = \tanh(-1.995) \approx -0.963. \quad (34)$$

Hence, one stochastic forward pass yields approximately 71.6% right steering, 82.1% throttle, and near-zero brake command (released brake channel).

Interpretation. The element-wise product $\odot$ controls exploration magnitude per actuator: small $\sigma_{a,i}$ damps stochasticity (e.g., gas), while larger $\sigma_{a,i}$ permits broader action perturbation (e.g., steering).

2 Phase 2: Backward Pass (Optimization Math)

$T = 4$: PER Memory Sampling (Experience Replay)

Transitions $\tau_i = (s, a, r, s')$ are sampled from prioritized replay with PMF:

$$P(i) = \frac{(|\delta_i| + \epsilon)^\alpha}{\sum_k (|\delta_k| + \epsilon)^\alpha}. \quad (35)$$

$T = 5$: RSSM Posterior State (Reality Alignment)

Posterior parameters conditioned on the true next observation:

$$\mu_{\text{post}}, \sigma_{\text{post}} = f_{\text{Encoder}}(h_t, F_{t+1}) \in \mathbb{R}^{1 \times 64}. \quad (36)$$

RSSM consistency objective:

$$\mathcal{L}_{\text{RSSM}} = D_{\text{KL}}(\mathcal{N}(\mu_{\text{post}}, \sigma_{\text{post}}^2) \parallel \mathcal{N}(\mu_{\text{prior}}, \sigma_{\text{prior}}^2)) \quad (37)$$

$$= \frac{1}{2} \sum_{j=1}^{64} \left(\log \frac{\sigma_{\text{prior},j}^2}{\sigma_{\text{post},j}^2} + \frac{\sigma_{\text{post},j}^2 + (\mu_{\text{post},j} - \mu_{\text{prior},j})^2}{\sigma_{\text{prior},j}^2} - 1 \right). \quad (38)$$

Directionality uses $D_{\text{KL}}(\text{posterior} \parallel \text{prior})$ as in variational inference. An optional KL-balancing variant is

$$\mathcal{L}_{\text{KL-bal}} = \eta D_{\text{KL}}(\text{sg}(q_{\text{post}}) \parallel p_{\text{prior}}) + (1 - \eta) D_{\text{KL}}(q_{\text{post}} \parallel \text{sg}(p_{\text{prior}})), \quad (39)$$

with stop-gradient operator $\text{sg}(\cdot)$ and, e.g., $\eta = 0.8$.

$T = 6$: MC-Dropout Variance (Epistemic Fear)

With dropout masks $M_k \sim \text{Bernoulli}(0.6)$ and $K = 5$ stochastic forward passes:

$$Q_k = f_{\text{Critic}}(F_t, z_t, a_t; M_k), \quad k \in \{1, \dots, 5\}. \quad (40)$$

Epistemic variance estimate:

$$\text{Var}_Q = \frac{1}{K-1} \sum_{k=1}^K (Q_k - \bar{Q})^2. \quad (41)$$

The denominator $K - 1$ is Bessel-corrected sample variance (important for small K , e.g., $K = 5$). The critic target includes a penalty term λVar_Q .

$T = 7$: Boltzmann Target Update (Algorithmic Entropy)

Soft actor-critic target:

$$y_t = r_t + \gamma \mathbb{E}_{s_{t+1} \sim p} \left[\min_{j=1,2} Q_{\theta'_j}(s_{t+1}, a_{t+1}) - \alpha \log \pi_{\phi}(a_{t+1} \mid s_{t+1}) \right]. \quad (42)$$

Actor gradient objective:

$$\nabla_{\phi} J_{\text{Actor}} = \nabla_{\phi} \mathbb{E}_{a \sim \pi} [\alpha \log \pi_{\phi}(a \mid s) - Q_{\theta}(s, a)]. \quad (43)$$

3 Thesis Application: Mathematical Proof of Deterministic Freeze

At deterministic evaluation, setting $\epsilon_a = 0$ at $T = 3$ gives

$$a_t = \tanh(\mu_a). \quad (44)$$

If MC-dropout at $T = 6$ yields high Var_Q , the critic value $Q_{\theta}(s, a)$ is reduced by uncertainty penalization. Then at $T = 7$, the actor objective pushes toward conservative mean actions. Under persistent high-variance penalties, optimization drives

$$\mu_a \rightarrow [0, 0, 0], \quad (45)$$

which induces low-throttle, low-steering behavior and can trap the agent near the start state.

4 Implementation Reinforcement: Theory vs. GPU Numerics

The theoretical equations above are analytically correct; however, practical training stability depends on numerically robust implementations. In particular:

1. the tanh-Jacobian correction should use the softplus-stable identity,
2. KL direction should remain $D_{\text{KL}}(\text{post}||\text{prior})$, with optional KL-balancing,
3. gating must be dimensionally aligned via explicit projection,
4. MC-dropout uncertainty should use Bessel-corrected variance.

Stating these implementation details in the methodology clarifies that the model is both mathematically rigorous and numerically executable.